

Problem Set for 24 April 2025

This exercise serves as an application of dual spaces in analysis wherein the elementary portion concerns the construction of a dual space known as the space of distributions or generalised functions thereby formalising the definition of the Dirac delta function and the latter optional part introduces the convolution operator and provides an elementary proof of Liouville's theorem for general harmonic functions as well as a relative version of the Stone-Weierstraß theorem applicable to a subspace defined by specific families of solutions to ordinary differential equations.

If one recognises the punctuation importance of the comma, one ought to employ it conscientiously in one's homework.

Exercise: Dual Spaces in Analysis Let $V = C_b^\infty(\mathbb{R}; \mathbb{R})$ denote the space of smooth, real-valued functions with compact support. This means that for any $f \in V$, the function f is smooth and satisfies $f(x) \equiv 0$ for all $|x| > x_0$, for some real number x_0 . The following problems concern the dual space V^* .

Note

In fact, the following problems holds for propositions holds for $V = C_b^\infty(\mathbb{R}^n; \mathbb{R})$.

1. (V is non-trivial) Demonstrate that V is not finite-dimensional.

Tip

Hint: Consider the function

$$a(x) := \begin{cases} e^{1/(x^2-1)}, & x \in (-1, 1), \\ 0, & \text{otherwise.} \end{cases}$$

It suffices to verify that $a^{(k)}(\pm 1) = 0$ for all k .

依数学归纳, $a^{(k)}(x) = q_k(x) \cdot \frac{a(x)}{(1-x^2)^k}$, 其中 $q_k(x)$ 是多项式. 依照数学分析, $\lim_{x \rightarrow \pm 1} \frac{a(x)}{(1-x^2)^k} = 0$.

2. (Differential Operator) Show that the operator $D : V \rightarrow V$, defined by $f \mapsto f'$, is well-defined. Determine the kernel $\ker D$, the image $\operatorname{im} D$, and the cokernel $\operatorname{coker} D$.

Caution

Remind that $f \in V$, and $f' \in V$.

若 $D(f) = 0$, 则 f 在任意闭区间上都是常数, 从而只能有 $f = 0$. 因此

- $\ker D = 0$.

$D(f)$ 的整体积分是 0. 同时, 整体积分为 0 的函数 g 给出 $G(x) := \int_{-\infty}^x g(t) \, dt \in V$, 满足 $DG = g$. 因此

- $\operatorname{im} D$ 是整体积分为 0 的函数.

整体积分是线性映射, 即,

$$I : V \rightarrow \mathbb{R}, \quad h \mapsto \int_{-\infty}^{\infty} h(t) \, dt.$$

因此

$$\bullet \operatorname{cok} D \simeq \frac{V}{\operatorname{im} D} = \frac{V}{\operatorname{ker} D} \simeq \operatorname{im} I \simeq \mathbb{R}.$$

3. (Integrable Functionals) Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is Riemann integrable over any bounded interval $[a, b]$. Show that the following map defines an element of V^* :

$$V \rightarrow \mathbb{R}, \quad f \mapsto \int_{-\infty}^{\infty} f(x)\varphi(x) \, dx.$$

① Note

It is more appropriate to work with $\mathcal{L}^1$ -space instead, as long as you are familiar with it.

只需验证积分收敛, 往后的加法和数乘是容易验证的. 不妨假定 f 在 $[a, b]$ 外均是 0, 且 $|f|$ 有上界
1. 此时, $\int_a^b f\varphi$ 的上下 Darboux 和之差不超过 $\int_a^b \varphi$ 者, 因此 Riemann 可积.

4. (V^* , merging some functions, and introducing some non-functions). The preceding exercise defines a mapping

$$\Phi : \{\text{Locally Riemann-Integrable Functions}\} \rightarrow V^*, \quad f \mapsto \left[g \mapsto \int_{-\infty}^{\infty} f(x)g(x) \, dx \right].$$

It is known that Φ is neither injective nor surjective.

- Let f be the function defined by $f(x) = 0$ for all $x \neq 1$, and $f(1) = 1$. Then $f \in \operatorname{ker} \Phi$.
- The Dirac delta functional δ , defined informally below, is not in the image of Φ . **This is left as an exercise for the reader.**

选用 T1 的函数, 记 $\mu(x) := \frac{a(x)}{e}$. 假定 $\delta = \Phi(\varphi)$, 则

$$1. \text{ 一方面 } (\delta \text{ 的定义}), \int_{\mathbb{R}} \mu(nx)\varphi(x) \, dx = \mu(0) = 1.$$

$$2. \text{ 另一方面, } \left| \int_{\mathbb{R}} \mu(nx)\varphi(x) \, dx \right| \leq \int_{-1/n}^{1/n} |\varphi(x)| \, dx \xrightarrow{n \rightarrow \infty} 0.$$

矛盾.

5. ($V \hookrightarrow V^*$) Show that $\operatorname{ker} \Phi \cap V = \{0\}$, and thus that V can be regarded as a subspace of V^* .

① Note

What if $f \in \ker \Phi$ with $\int_a^b f \neq 0$? How we find $\varphi \in V$ such that $\int_{-\infty}^{\infty} f\varphi \neq 0$?

$\int_{\mathbb{R}} f \cdot \Phi(f) dx \geq 0$, 取等当且仅当 $f = 0$. 因此 $\Phi(f) = 0$ 当且仅当 $f = 0$.

6. (The Dirac Delta Functional) The Dirac delta functional δ is informally described as:

$$\delta(x) = \begin{cases} 0, & x \neq 0, \\ \infty, & x = 0, \end{cases} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1.$$

Provide a formal definition of δ as an element of V^* .

Note

We call δ a distribution, a generalised function, a functional on V , or whatever you like.

δ 是对偶空间 V^* 中的元素, 定义作

$$\delta : V \rightarrow \mathbb{R}, \quad f \mapsto f(0).$$

7. (Generalised Derivatives) Define φ as a piecewise linear function passing through the points:

$$(-\infty, 0) \rightarrow (-1, 0) \rightarrow (0, 2) \rightarrow (1, -1) \rightarrow (2, 0) \rightarrow (+\infty, 0).$$

Although φ is continuous, its derivative φ' is not classically defined. Use the identity

$$\int_{-\infty}^{\infty} f(x)g'(x) dx = - \int_{-\infty}^{\infty} f'(x)g(x) dx$$

to define φ' in the distributional sense. Express φ'' explicitly as a linear combination of shifted Dirac delta functions $\delta(x - a)$.

Tip

Try firstly some simple examples on your first trial, i.e., how would you like define the derivation of sign function $\text{sgn}(x)$? The answer is $\text{sgn}' = 2\delta$.

Note

$\varphi \in C^0(\mathbb{R}; \mathbb{R})$ has no derivatives, but $\varphi \in V^*$ has formal derivatives in V^* .

对任意 $f \in V$,

$$\int_{\mathbb{R}} f(x)\varphi''(x) \, dx := \int_{\mathbb{R}} f''(x)\varphi(x) \, dx.$$

分段计算,

1. $\int_{-1}^0 2(x+1)f''(x) \, dx = 2f(-1) - 2f(0) + 2f'(0).$
2. $\int_0^1 (2-3x)f''(x) \, dx = -3f(0) + 3f(1) - 2f'(0) - f'(1).$
3. $\int_1^2 (x-2)f''(x) \, dx = f(1) - f(2) + f'(1).$

相加, 得

$$\int_{\mathbb{R}} f''(x)\varphi(x) \, dx := 2f(-1) - 5f(0) + 4f(1) - f(2).$$

因此, 在 V^* 中定义

$$\varphi''(x) : 2\delta(x+1) - 5\delta(x) + 4\delta(x-1) - \delta(x-2).$$

当然, 这一答案也能猜到, 例如 φ' 是阶梯函数, 二阶导数是阶梯高度差, 用 δ 表示即可.

8. (Generalised Limits) For each $\varphi \in V \hookrightarrow V^*$, define a sequence of functions for each $n \in \mathbb{N}_+$ by:

$$(-)_n : V \rightarrow V, \quad \varphi \mapsto [x \mapsto n \cdot \varphi(nx)].$$

It is evident that:

$$\int_{-\infty}^{\infty} \varphi_n(x) \, dx = \int_{-\infty}^{\infty} \varphi(x) \, dx.$$

The sequence $\{\varphi_n\}$ does not converge uniformly in V . However, show that there exists $L \in V^*$ such that:

$$L(f) = \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f(x)\varphi_n(x) \, dx.$$

Your task is to show that the limit exists for arbitrary f , and that L is linear.

⚠ Caution

Linear mappings do not, in general, preserve infinite sums.

📌 Note

$\{\varphi_n\}$ are neither uniformly convergent, nor pointwise convergent. There exists some new types of convergence in V^* .

改变参考系. 记 $f^n(x) := f(\frac{x}{n})$, 此时

$$\int_{-\infty}^{\infty} f^n(x) \varphi(x) \, dx = \int_{-\infty}^{\infty} f(x) \varphi_n(x) \, dx.$$

不妨设 φ 仅在 $[-1, 1]$ 区间外恒零. 由 0 点处的连续性, f^n 在 $[-1, 1]$ 一致收敛至常函数 $f(0)$. 因此

$$L(f) = f(0) \cdot \int_{\mathbb{R}} \varphi(x) \, dx.$$

依照定义, 这是线性的.

Optional Problems: Another Structure on $\text{End}(V)$

1. **(Convolution Operator)** Show that the following map is a linear injection:

$$\star : V \rightarrow \text{End}(V), \quad f \mapsto \left[g \mapsto \left(x \mapsto \int_{-\infty}^{\infty} f(y)g(x-y) \, dy \right) \right].$$

Note

By Currying $(V \rightarrow (V \rightarrow V)) \simeq (V \times V \rightarrow V)$, the operation $\star$ is also a bilinear map of the type

$$V \times V \rightarrow V, \quad (f, g) \mapsto \left[x \mapsto \int_{-\infty}^{\infty} f(y)g(x-y) \, dy \right]$$

2. **(Multiplicative Structure of V)** Since $V \subseteq \text{End}(V)$ and $V \star V \subseteq V$, the space V acquires a multiplicative structure compatible with its $\mathbb{R}$ -linear structure. Hence, V forms a ring without unit. (The identity element should be $\delta(x)$.)

- The ring $(\text{End}(V), +, \star)$ lacks a unit, while $(\text{End}(V), +, \circ)$ possesses the unit id_V .

Show that $(\text{End}(V), +, \star)$ is commutative: that is, $f \star g = g \star f$.

Note

$\text{End}(V)$ is both a ring and an $\mathbb{R}$ -linear space where the Abelian group parts coincides. We call $\text{End}(V)$ an $\mathbb{R}$ -algebra.

3. **(Derivations and Convolutions)** The ring $(\text{End}(V), +, \circ)$ complicates differentiation due to the chain rule. However, convolution satisfies:

$$(f \star g)' = f' \star g = f \star g'.$$

Note

The relationship between $\circ$ and $\star$ is analogous to that between matrix multiplication and the Hadamard product.

4. **(Convolution with Locally Integrable Functions)** Show that

$$\{\text{Locally Riemann-Integrable Functions}\} \star V$$

is a linear subspace of the space of smooth functions.

Important

The problems surrounding $V \star V \stackrel{?}{=} V$ are known as *Ehrenpreis factorisation problems*.

For $V = C_b^\infty(\mathbb{R}^n; \mathbb{R})$, the case $n = 1$ admits a positive answer.

5. (Mean Value Property) A function f satisfies the *Mean Value Property* (MVP) if:

$$f(x) = \frac{1}{2a} \int_{x-a}^{x+a} f(t) dt$$

for all x and a . This property is trivial in one variable but non-trivial in higher dimensions (cf. [harmonic functions](#), especially the real and imaginary parts of [holomorphic functions](#)). Let

$$a(x) := e^{1/(x^2-1)} \quad \text{or} \quad a(\mathbf{x}) := e^{1/(\mathbf{x}^T \mathbf{x} - 1)}.$$

Show that $a \star f = C \cdot f$ for some constant C , whenever f satisfies MVP. Determine the value of C . Demonstrate that MVP functions are smooth.

Important

Let $p(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$ be radially symmetric, which means that p factors through the Euclidean norm $\mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{x} \mapsto \|\mathbf{x}\|$. Now any MVP function is an eigenvector of $p \star (-)$ with eigenvalue $\int_{\mathbb{R}^n} p$.

6. (Locally MVP Functions) A function f is *locally MVP* if:

$$f(x) = \frac{1}{2a} \int_{x-a}^{x+a} f(t) dt$$

for all x and $a \in (0, N(x))$, where $N(x)$ is continuous. Show that locally MVP functions are smooth.

Tip

A function is smooth iff it is smooth at every points. Apply **3.** to transfer the derivatives onto a .

7. (Liouville's Theorem) Prove that any bounded locally MVP function defined on $\mathbb{R}^n$ taking values in $[a, b]$ must be constant.

Tip

To show that $f(p_1) = f(p_2)$ for arbitrary $p_i \in \mathbb{R}^n$, one can take the balls

$$B(p_i, R) := \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - p_i\| \leq R\}.$$

Now show that $\lim_{R \rightarrow \infty} \frac{\int_{B(p_i, R) \cap B(p_2, R)} f(\mathbf{x}) d\mathbf{x}}{\text{Volume}(B(0, R))}$ is both $f(p_1)$ and $f(p_2)$. The volume of $B(p_1, R) \setminus B(p_2, R)$ is controlled by R^{n-1} .

Important

By magic property of $a \star f = C \cdot f$, we prove that

1. the local MVP-property implies smoothness,
2. Liouville's theorem.

If you are interested in analysis, show that being annihilated by the Laplacian

$\Delta := \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$ is equivalent to locally MVP property (we call these harmonic functions).

The derivatives of harmonic functions is also harmonic, since

$$\Delta D^{(?) } f = (\Delta f) \star (D^{(?) } a) = 0.$$

8. (Smooth Approximation via Convolution) Let f be any continuous function and let $g \in V$ satisfy $\int g = 1$. Show that the sequence $(f \star g_n)_{n \geq 1}$ converges uniformly to f on any bounded interval, where $(-)_n$ is defined as in Exercise 1.8.

📌 Important

How to construct a smooth function which approximates (a relative conception) an ugly function? Try $(-) \star g_n$ for n large enough!

9. (Stone-Weierstraß Theorem) Let f be any continuous function. Show that there exists a sequence of polynomials converging uniformly to f on any bounded interval. You may assume $f^{-1}(0) \supseteq \mathbb{R} \setminus [-1, 1]$.

- Let P_n be the space of polynomials of degree at most n . Show that $V \star P_n \subseteq P_n$, and thus $V \star \mathbb{R}[x] \subseteq \mathbb{R}[x]$. In other word, $\mathbb{R}[x]$ is an invariant subspace of the operator $f \star (-)$.
- For each $n \geq 1$, there exists a polynomial p_n such that $\int_{-n}^n p_n(x) dx = 1$ and $\int_{-1/n}^{1/n} p_n(x) dx > 1 - n^{-114514}$.
- Show that $f \star p_n$ converges uniformly to f on the interval $[-1, 1]$.

💡 Tip

For the first step, show that $f \in V$ with $f'(x) \equiv 0$ is a linear function with degree 1.

For the second step, construct from $(1 - (nx)^2)^m$.

10. (Relative Stone-Weierstraß Theorem) An analogue of the Stone-Weierstraß theorem is established using trigonometric functions. Let T_n denote the space of trigonometric polynomials of degree at most n . Show that $V \star T_n \subseteq T_n$, noting that functions like $t \star f$ and t satisfy the same ordinary differential equations. For instance:

$$(\sin \star f)'' + \sin \star f = (\sin'' + \sin) \star f = 0,$$

so $\sin \star f$ is of the form $A \sin(x + \beta)$. Use this to establish the T -version of the Stone-Weierstraß theorem.

Note

For trigonometric approximations, it is much easier to construct with Dirichlet kernel or Fejér kernel.